

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
	(b)	(i)	Recall of $I = P \div 4\pi R^2$ in any form (1) Substitution: P (or Luminosity) = $1.32 \times 10^{-8} \times 4\pi \times (1.58 \times 10^{17})^2$ (1) $P = 4.1(4) \times 10^{27}$ W seen (1)	1	1 1		3	3	
		(ii)	Substitution: $P = A\sigma T^4$ in any form e.g.: $4 \times 10^{27} = A \times 5.67 \times 10^{-8} \times (7700)^4$ (1) [Accept $4.1(4)10^{27}$ for P] $A = 2.0 \times 10^{19} \text{ m}^2$ (1) [2.1×10^{19} if 4.14×10^{27} used] Diameter = $2.5 \times 10^9 \text{ m}$ (1) [2.6×10^9 if 2.1×10^{19} used] Use of πR^2 rather than $4\pi R^2 \rightarrow 5 \times 10^9 \text{ m}$ (2 marks)	1 1	1		3	3	
			Question 6 total	9	3	0	12	6	0